FIT5201 - Machine Learning

Online Quiz 1 (2.25%) opens this Week

Deadline: 11:55PM on Tue of Week 3

FIT5201 - Machine Learning

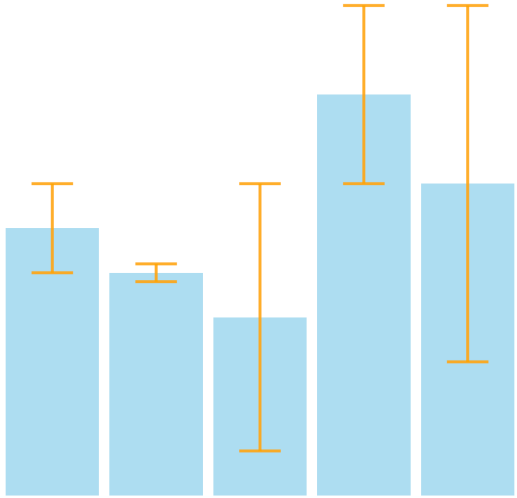
Module 1: Elements of Machine Learning

Part B:

- **Probabilistic Machine Learning**
  - ☐ basics of probability theory
  - ☐ probabilistic models and parameter estimation
  - ☐ quantifying uncertainty about arbitrary estimates (bootstrap)
  - ☐ Bayesian approach to ML

Uncertainty in Machine Learning (ML)

Intuition:
Cannot base decision on prediction without quantifying uncertainty

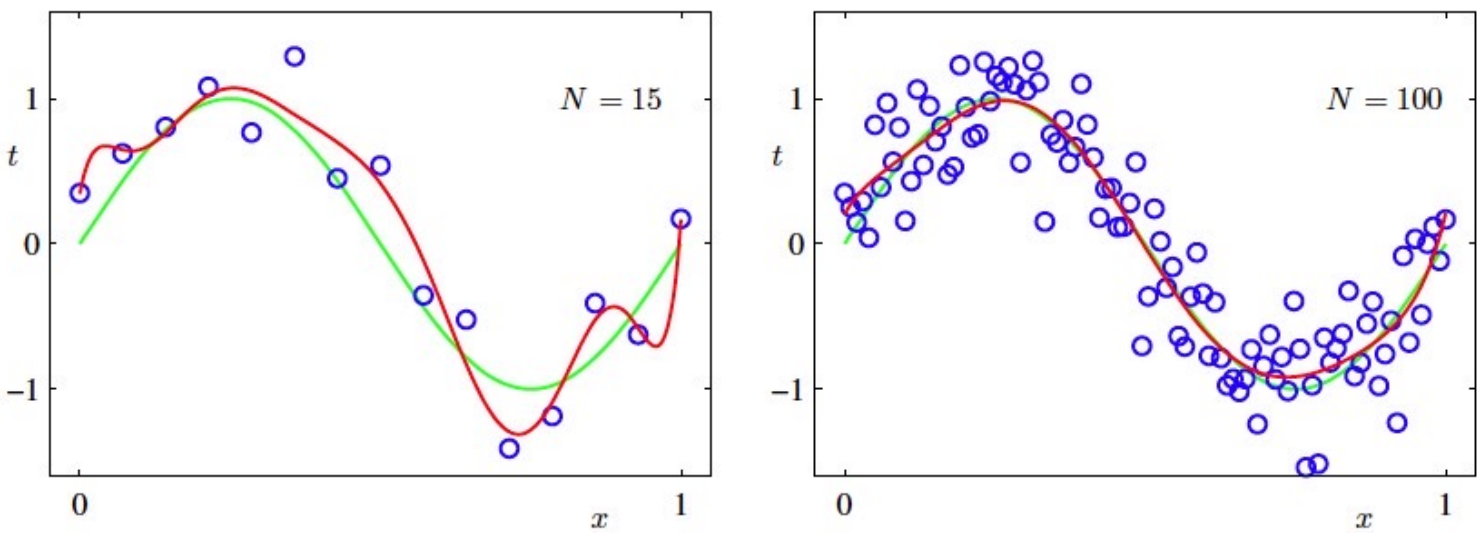


What is uncertainty and where does it come from?

Uncertainty in Machine Learning (ML)

Non-determinism of modelled process (e.g., measurement noise): aleatoric uncertainty

Finite size of (random) training data / limited knowledge: epistemic uncertainty



Probability Theory

- Provides framework for defining and manipulating "uncertainty"

- Used in ML to
  - Define learning task in a meaningful way (generalisation error, last week)
  - Enrich our concept of "model" from predictive to probabilistic
  - Quantify uncertainty about learned model parameters and predictions

Appendix A in Week 1 / Tutorial Resources / "**Module one.pdf**"

Basics of Probability Theory

Mathematical model of (repeatable) random experiments like:

flipping a coin

rolling two dice

athlete performing long jump

Random Variables

Random variables ”measure” specific aspects of experiment outcome like:

side in which coin lands

sum of eyes rolled

distance jumped

Random Variables

Domain of random variable is associated set of possible values

$\{\mathrm{H(ead)}, \mathrm{T(ail)}\}$

$\{1, \ldots, 12\}$

$\mathbb{R}_+$

Distributions

Abstract notation

- Random variable X (capital letter)
- Domain D_X
- Concrete value $x \in D_X$ (non-capital letter)
- Event is subset $\mathcal{X} \subseteq D_X$ (e.g., even dice sum: $\mathcal{X} = \{2, 4, \ldots, 10, 12\}$)

Probability distributions

In limit of total number of trials (n) to infinity, can define the probability of an event as limit of fraction:

$$\mathrm{P}(X \in \mathcal{X}) = \lim_{n \to \infty} \frac{\#\text{trials with } X \in \mathcal{X}}{n}$$

Set of all those probabilities is called the probability distribution of X.

Discrete Random Variables

RV is called discrete if there is probability mass function $p_X(x): D_X \to \mathbb{R}_+$ such that for all events $\mathcal{X} \subseteq D_X$

$$\mathrm{P}(X \in \mathcal{X}) = \sum_{x \in \mathcal{X}} p_X(x)$$

Example: sum of two dice

X	p
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36

$$\begin{aligned}\mathrm{P}(X \in \{2,4,\dots,12\}) &= p_X(2) + p_X(4) + \cdots + p_X(12) \\ &= \frac{1}{36} + \frac{3}{36} + \frac{5}{36} + \frac{5}{36} + \frac{3}{36} + \frac{1}{36} \\ &= \frac{18}{36} = \frac{1}{2}\end{aligned}$$

Discrete Random Variables

RV is called discrete if there is probability mass function $p_X(x): D_X \to \mathbb{R}_+$ such that for all events $\mathcal{X} \subseteq D_X$

$$\mathrm{P}(X \in \mathcal{X}) = \sum_{x \in \mathcal{X}} p_X(x)$$

Observations / axioms:

1. Probabilities are bounded, i.e., for all x in D_X, $p_X(x) \in [0, 1]$
2. Probabilities sum to 1, i.e.,

$$\sum_{x \in D_X} p_X(x) = 1$$

Convention:

We sometimes drop subscript X from p and domain from sums *when clear from context, e.g.,*

$$\sum_{x} p(x) = 1$$

Two or More Variables: Joint and Marginal Probabilities

Example 2: Tossing two coins

Random Variables: X, Y, Z

Define three random variables

X: tossing the coin for the first time ($D_X = \{H, T\}$)
Y: tossing the coin for second time ($D_Y = \{H, T\}$)
Z: number of head outcomes ($D_Z = \{0, 1, 2\}$)

Joint probability (mass function) $p(\cdot,\cdot,\cdot)$ gives probability of possible combination of values:

X	Y	Z	p
T	T	0	$4/9$
T	H	1	$2/9$
H	T	1	$2/9$
H	H	2	$1/9$

(omitted entries w. $p = 0$)

marginal probability

$$p_X(x) = \sum_{y,z} p(x, y, z)$$

X	p
H	$1/3$
T	$2/3$

Again, all probabilities sum to 1: $\sum_{x,y,z} p(x, y, z) = 1$

Two or More Variables: Joint and Marginal Probabilities

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marginal probability

$$p_Z(z) = \sum_{x,y} p(x, y, z)$$

Z	p
0	$4/9$
1	$4/9$
2	$1/9$

Again, all probabilities sum to 1: $\sum_{x,y,z} p(x, y, z) = 1$

Conditional Probability

The conditional probability of $X = x$ *given* $Y = y$ for $x \in D_X$ and $y \in D_Y$ with $p_Y(y) > 0$:

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

Double Coin Toss Example

X: tossing the coin for the first time ($D_X = \{H, T\}$)
Y: tossing the coin for second time ($D_Y = \{H, T\}$)
Z: number of head outcomes ($D_Z = \{0, 1, 2\}$)

What is the probability of two heads in total given that the first toss comes up heads?

$$p_{Z|X}(2 \mid H) = \frac{p_{Z,X}(2, H)}{p_X(H)} = \frac{1}{3}$$

Conditional Probability

The conditional probability of $X = x$ *given* $Y = y$ for $x \in D_X$ and $y \in D_Y$ with $p_Y(y) > 0$:

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

Product rule (chain rule)

$$p(x, y) = p(x|y)p(y)$$

Bayes' theorem

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}$$

Example task: compute probability of first coin "heads" given that total of one "head"

$$p_{X|Z}(H|1) = \frac{p_{Z|X}(1 \mid H)p_X(H)}{p_Z(1)} = \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{4}{9}} = \frac{1}{2}$$

Two Variables: Independence

Two variables X and Y are independent of for all $x \in D_X$ and $y \in D_Y$:

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

or alternatively if for all $x \in D_X$ and $y \in D_Y$ with $p(y) > 0$:

$$p(x) = p(x|y)$$

Double Coin Toss Example

X: tossing the coin for the first time ($D_X = \{H, T\}$)
Y: tossing the coin for second time ($D_Y = \{H, T\}$)
Z: number of head outcomes ($D_Z = \{0, 1, 2\}$)

X and Y are independent

X and Z are dependent

$$0 = p_{X,Z}(H, 0) \neq p_X(H)p_Z(0) = 4/27$$

Continuous Random Variables

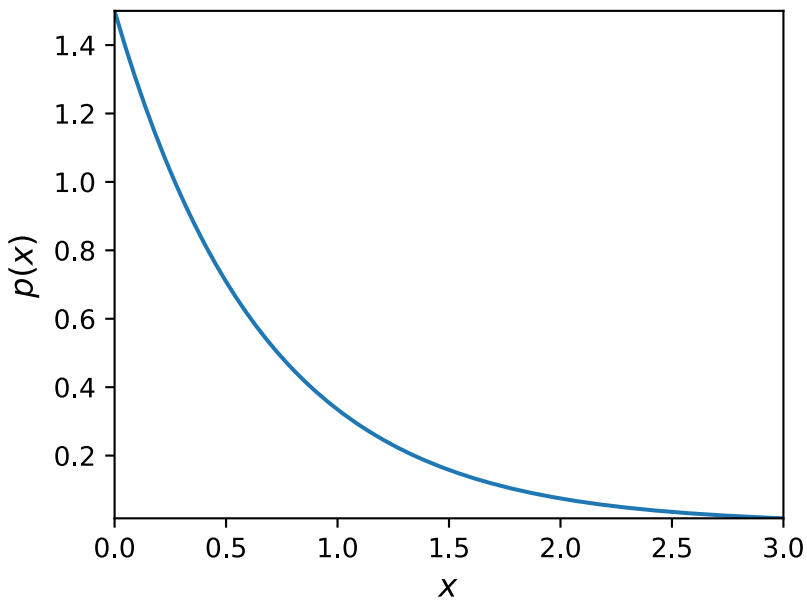
Continuous random variables X with domain $D_X \subseteq \mathbb{R}$ and $P(X = x) = 0$

Distribution given by **probability density function (pdf)** $p(x) \geq 0$ that describes probability of event $a \leq X \leq b$ by integration

$$P(a \leq X \leq b) = \int_a^b p(x)dx$$

Example: time until you receive first email on random workday

$$p(x) = \frac{3}{2}\exp\left(-\frac{3}{2}x\right)$$



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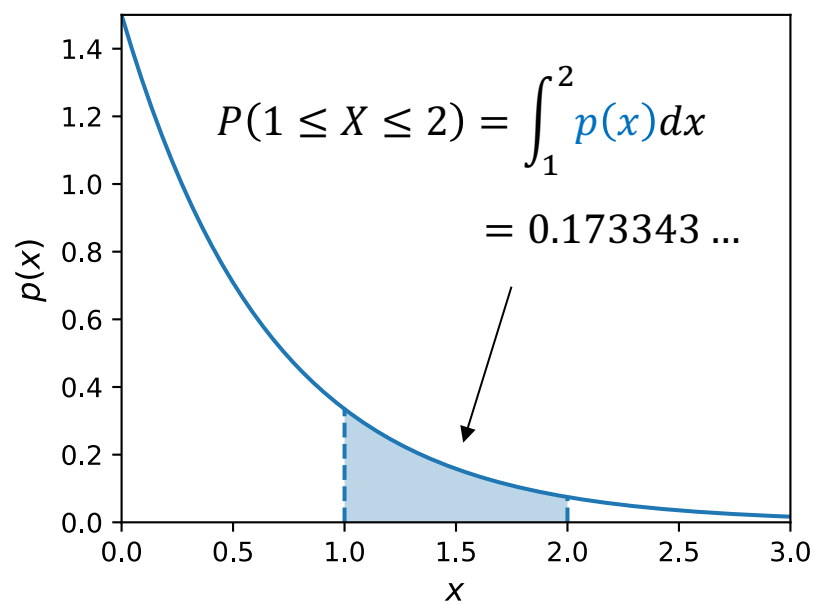
Example: time until you receive first email on random workday

$$p(x) = \frac{3}{2}\exp\left(-\frac{3}{2}x\right)$$

Question: probability of receiving first mail in second hour of work?



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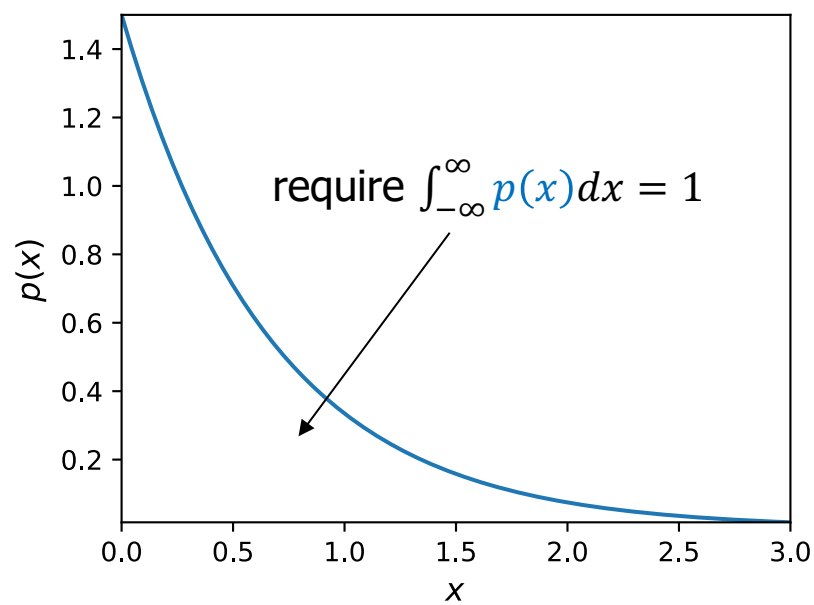

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Example: time until you receive first email on random workday

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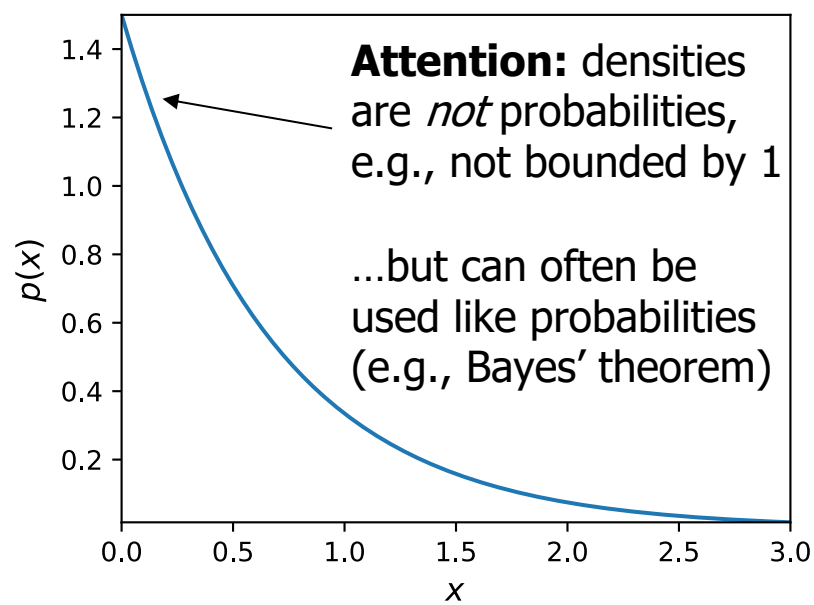


Expected (Mean) Values

If $D_X \subseteq \mathbb{R}$, we define expected value $\mathbf{E}[X]$ of X as:

$$\mathbf{E}[X] = \int_{D_X} x\, p(x)\, dx$$

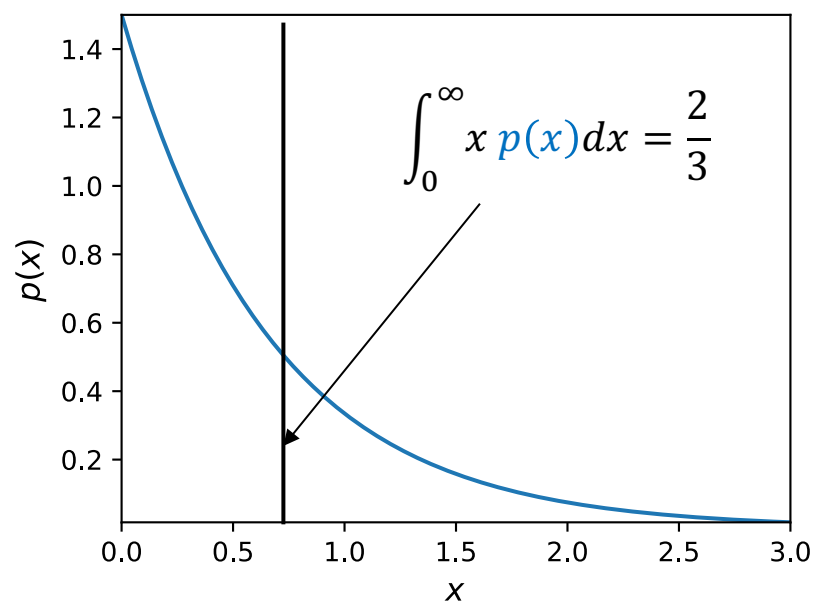
continuous X

$$\mathbf{E}[X] = \sum_{x \in D_x} x\, p(x)$$

discrete X

Question: when do I receive my first email "on average"

$$p(x) = \frac{3}{2} \exp\left(-\frac{3}{2} x\right)$$



See Activity 0.2 for more details

Expected (Mean) Values

If $D_X \subseteq \mathbb{R}$, we define expected value $\mathbf{E}[X]$ of X as:

$$\mathbf{E}[X] = \int_{D_X} x\, p(x)\, dx$$

continuous X

$$\mathbf{E}[X] = \sum_{x \in D_x} x\, p(x)$$

discrete X

Question: What is the average number of "head" tosses

Z	p
0	4/9
1	4/9
2	1/9

$$\mathbf{E}[Z] = 0 \cdot \frac{4}{9} + 1 \cdot \frac{4}{9} + 2 \cdot \frac{1}{9} = \frac{2}{3}$$

Conditional Expectation

For $y \in D_Y$ with $p(y) > 0$, we can also define conditional expectation:

$$\mathbf{E}[X \mid Y = y] = \int_{D_X} x\, p(x \mid y)\, dx \qquad \mathbf{E}[X \mid Y = y] = \sum_{x \in D_x} x\, p(x \mid y)$$

continuous X

discrete X

Question: what is the expected number of "head" outcomes given that the first toss is "tail" / "head":

$$\mathbf{E}[Z \mid X = T] = 0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} + 2 \cdot 0 = \frac{1}{3}$$

$$\mathbf{E}[Z \mid X = H] = 0 \cdot 0 + 1 \cdot \frac{2}{3} + 2 \cdot \frac{1}{3} = \frac{4}{3}$$

X	Y	Z	p
T	T	0	$4/9$
T	H	1	$2/9$
H	T	1	$2/9$
H	H	2	$1/9$

Probabilistic Models: Learning about a coin

Probabilistic Models

Last week: introduced predictive models, i.e., parameterised prediction functions

$$y(x, w)$$

Now: probabilistic models, (conditional) probability distributions that (again) depend on parameters w (and features x):

$$p(y|x, w)$$

Motivation:

- ❑ naturally also provides prediction $\mathbf{E}[Y|X = x, w]$
- ❑ but also level of certainty

Probabilistic Models

A **probabilistic model** is simply a (conditional) probability distribution that depends on parameters w (and features x):

$$p(y|x, w)$$

Example: assume dataset with outcomes of 10 independent tosses of the same coin

$$\boldsymbol{D} = \{\boldsymbol{H}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{H}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{H}, \boldsymbol{T}\}$$

Task: build a model for this coin tossing to estimate the probability that next coin toss will be a head (or tail).

Learning about a coin

Example: assume dataset with outcomes of 10 independent tosses of the same coin

$$\boldsymbol{D} = \{\boldsymbol{H}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{H}, \boldsymbol{T}, \boldsymbol{T}, \boldsymbol{H}, \boldsymbol{T}\}$$

Our **model** has one parameter $w \in [0, 1]$ that models probability of head:

$$p(y|w) = \begin{cases} w, & y = H \\ 1 - w, & y = T \end{cases}$$

Problem: need to estimate w from training data D.

Maximum Likelihood Principle

Idea: maximise probability of data as function of parameters w

$$p(D|w)$$

This is called the **likelihood function** and maximiser w_{ML} is called maximum likelihood estimate:

$$p(D|w_{\mathrm{ML}}) = \max_{w} p(D|w)$$

With independent data $D = \{y_1, \dots, y_n\}$ likelhood becomes product

$$p(D|w) = p(y_1|w)p(y_2|w) \dots p(y_n|w) = \prod_{i=1}^{n} p(y_i|w)$$

Likelihood Function for Coin Example

Data

$$D = \{H, T, T, T, T, H, T, T, H, T\}$$

Probability for individual data point

$$p(y|w) = \begin{cases} w, & y = H \\ 1 - w, & y = T \end{cases}$$

Likelihood

$$\begin{aligned} P(D|w) &= w(1-w)(1-w)(1-w)(1-w)w(1-w)(1-w)w(1-w) \\ &= w^3(1-w)^7 \\ &= w^{|H|}(1-w)^{|T|} \end{aligned}$$

Maximum Likelihood

- **D = {H, T, T, T, T, H, T, T, H, T}**

$$p(D|w) = w^3(1-w)^7$$

If w had been either 0 (heads impossible) or 1 (tails impossible), then the probability of getting D would have been 0.

If w had been ½ (a fair coin), then the probability of getting D would have been 1/1024.

Find Maximum Likelihood Estimate

Idea: use derivative criterion to identify max of likelihood function

$$\frac{\partial}{\partial w} p(D|w)\bigg|_{w=w_{\mathrm{ML}}} = 0$$

Problem: product very hard to differentiate

Instead: consider log likelihood function (has same maximiser)

$$\ln p(D|w) = \ln \prod_{i=1}^{n} p(y_i|w) = \sum_{i=1}^{n} \ln p(y_i|w)$$

and work with

$$\frac{\partial}{\partial w} \ln p(D|w) = \sum_{i=1}^{n} \frac{\partial}{\partial w} \ln p(y_i|w)$$

MLE for Coin Example

Data

$$D = \{H, T, T, T, T, H, T, T, H, T\}$$

Likelihood

$$p(D|w) = w^3(1-w)^7$$

Log likelihood

$$\ln p(D|w) = 3\ln w + 7\ln(1-w)$$

Derivative criterion

$$\frac{\partial}{\partial w}\ln p(D|w) = \frac{3}{w} - \frac{7}{1-w} = 0$$

Result: $w_{\mathrm{ML}} = 0.3$

MLE for Coin Example in General

Likelihood

$$p(D|w) = w^{|H|}(1-w)^{|T|}$$

Log likelihood

$$\ln p(D|w) = |H| \ln w + |T| \ln(1-w)$$

Derivative criterion

$$\frac{\partial}{\partial w} \ln p(D|w) = \frac{|H|}{w} - \frac{|T|}{1-w} = 0$$

Maximum likelihood estimate

$$w_{\mathrm{ML}} = \frac{|H|}{|H| + |T|}$$

Learning about a coin

❑ Summary

- The prediction problem (probability that next coin is H or T) is formulated as finding an optimal coin toss parameter w that maxmises the probability that we observe D
- To find such w, we can use logarithm and derivative calculation
- Once we obtain such w, we can use it for prediction purposes.
- The parameter w clearly depends on the dataset D. So w can be changing if we have a different dataset D'.

Bootstrap

Uncertainty Quantification

By moving from predictive to probabilistic model we acknowledge that our target is not necessarily a deterministic quantity (given the input)

Aleatoric uncertainty: inherent *non-determinism* of modelled process

Example: uncertainty of outcome of coin flip even if exact probability known

However, there is an additional source of uncertainty due to the finitude of our specific observed dataset

Epistemic (data) uncertainty: lack of *knowledge* about the process

Example: uncertainty about probability of heads

Parameter Uncertainty Conceptually

- We had a specific (random) dataset D to find parameters $\boldsymbol{w}(D)$, e.g., with maximum likelihood

- **Observation:** that makes $\boldsymbol{w}$ as well as individual parameters w a random variable with some distribution

- **Conceptually:** the less the distribution of w is concentrated, the more uncertain we are about our specific estimate $w(D)$

- **Idea:** if we could sample new datasets, we could start to estimate that distribution (and the dispersion / variance of w)

- **Problem:** in practice we have only one dataset, and we would like to use all data for learning w

Bootstrap for Quantifying Uncertainty

Now, the problem is how to get the alternate datasets?

clue: generate the alternate datasets from the original dataset. But how?

Solution: resampling with bootstrap strategy

Basic idea of bootstrap: approximate the real distribution in the infinity situation with the empirical distribution

Bootstrap Example

A bootstrap sample is a random sample conducted with replacement

1. Randomly select an observation from the original data
2. Write it down
3. Put it back (i.e. Any observation can be selected more than once)

Repeat these steps 1-3 N times; N is the number of observations in the original sample

Final Result: One "bootstrap sample" with N observations

0	38	1

Original Data

N

Target	Age	Inquires
0	48	3
0	37	1
0	37	1
	…	
0	24	5

Summary

Classical or frequentist interpretation of uncertainty

1. View probabilities in terms of the frequencies of random, repeatable events, in the limit of infinity.

2. Use bootstrap sample strategy to simulate infinity based on the available data.

3. Learn the model by maximizing the likelihood of every single bootstrap data set.

4. Evaluate the uncertainty by looking at the variability of predictions between the different bootstrap data sets.

Bayesian Approach to Machine Learning

(use probabilities to quantify the uncertainty directly)

Motivation for Bayesian Approach

- What's the limitation of the above model?
  - Imagine that **D = {T, T}** with the same coin! Then, what's the probability that the next coin toss will be a head?

$$\mathbf{w}_{\mathbf{MLE}} = \frac{|H|}{|H| + |T|} = \frac{\mathbf{0}}{\mathbf{2} + \mathbf{0}} = \mathbf{0}$$

  - The result $w = 0$ is unreasonable.
  - Fortunately, from your experience, most coins are about 50-50.
  - Use **Bayesian modelling** to encode this **prior experience or knowledge about the parameter:** $p(w)$

What is Bayesian Approach to ML

1. Define probabilistic model (likelihood) $p(D|\boldsymbol{w})$ as before

2. Assume parameters $\boldsymbol{w}$ of model $p(y|\boldsymbol{x}, \boldsymbol{w})$ are itself a random variable

3. Define marginal distribution $p(\boldsymbol{w})$, called **prior distribution**, that captures prior knowledge / belief about parameter

4. Compute **posterior distribution** $p(\boldsymbol{w}|D)$ with Bayes' Theorem

$$p(\boldsymbol{w}|D) = \frac{p(D|\boldsymbol{w})p(\boldsymbol{w})}{p(D)}$$

posterior likelihood prior

This directly gives us uncertainty about parameters!

Encoding Prior Knowledge

The probablity distributions $p(w)$ follow Beta distribution,

$$\mathbf{Beta}(\boldsymbol{w}|\boldsymbol{a},\boldsymbol{b}) \propto \boldsymbol{w}^{\boldsymbol{a}-\mathbf{1}}(\mathbf{1}-\boldsymbol{w})^{\boldsymbol{b}-\mathbf{1}}, \boldsymbol{w} \in [\mathbf{0},\mathbf{1}]$$

We have seen this before: $\boldsymbol{w}^{\boldsymbol{a}-\mathbf{1}}(\mathbf{1}-\boldsymbol{w})^{\boldsymbol{b}-\mathbf{1}}$ (very similar to the estimate in MLE). Remember: $\boldsymbol{w}^{\mathbf{3}}(\mathbf{1}-\boldsymbol{w})^{\mathbf{7}}$

In the MLE estimate, we had a similar function form but (a-1) and (b-1) were substituted by the number of Heads and Tails respectively!

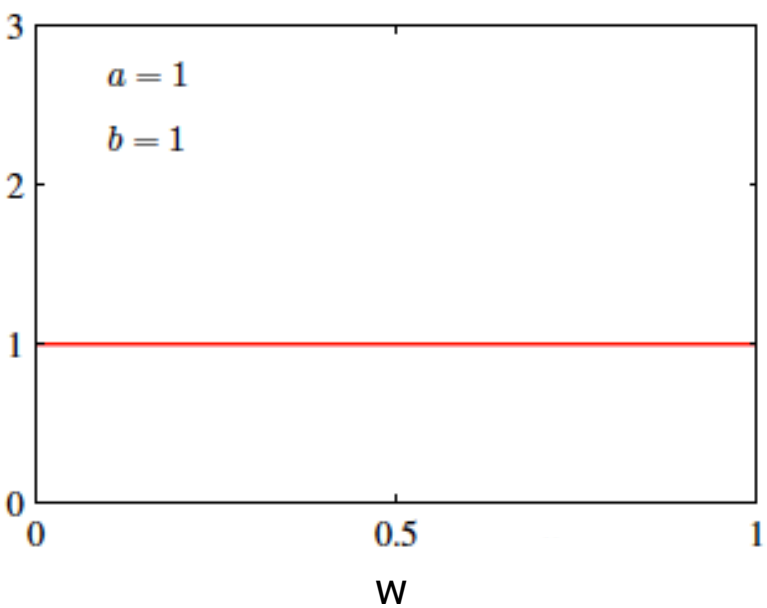
Highlight that Beta distribution encodes prior knowledge about parameter w **before** actually observing data.

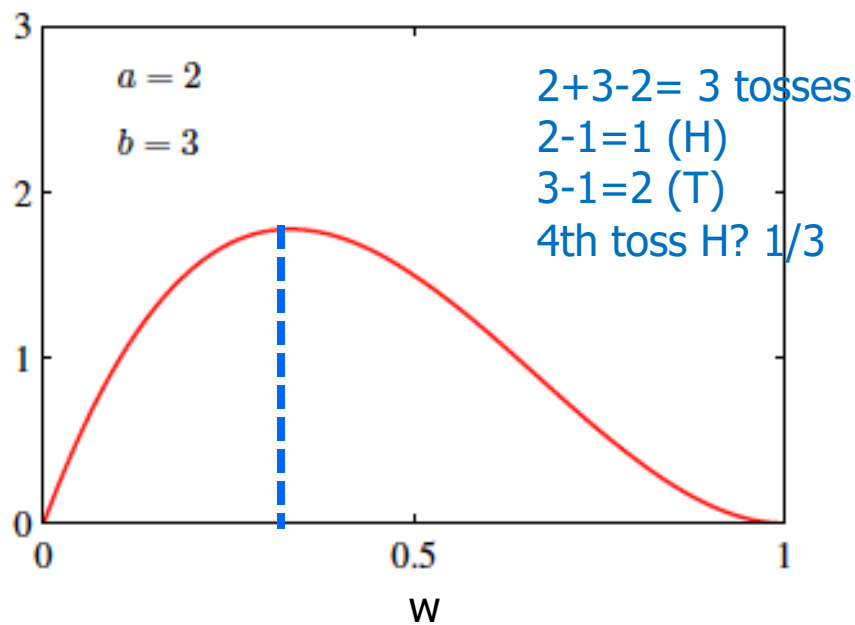
Encoding Prior Knowledge

Prior knowledge (or belief),

$$\mathbf{p}(\mathbf{w}) \propto \mathbf{Beta}(\boldsymbol{w}|\boldsymbol{a},\boldsymbol{b}) \propto \boldsymbol{w}^{\boldsymbol{a}-\mathbf{1}}(\mathbf{1}-\boldsymbol{w})^{\boldsymbol{b}-\mathbf{1}}, \boldsymbol{w} \in [\mathbf{0},\mathbf{1}]$$

How to interprete: (a-1) and (b-1) are the number of Heads and Tails I think we would see, if we made (a+b-2) coin tosses!





What is Bayesian Approach to ML

For any x, I want to estimate $p(w = x|D)$, where D is a randome variable that models the observed data: simply we denote this $p(w|D)$.

Bayes' Theorem:

$$p(w|D) = \frac{p(D|w)p(w)}{p(D)}$$

- $p(D|w)$: We know this one! This is used to estimate parameter w in MLE
- $p(w)$: We know this one too! This is prior belief – probability distribution over parameter w
- $p(D)$: that's the probability of the observed data D

What is Bayesian Approach to ML

Bayes' Theorem:

$$p(w|D) = \frac{p(D|w)p(w)}{p(D)}$$

Let's put together the numerator!

$$p(D|w) = w^{|H|}(1-w)^{|T|} \text{ (by MLE)}$$
$$p(w) \propto w^{a-1}(1-w)^{b-1} \text{ (by prior belief)}$$

So we get the numerator:

$$p(w|D) \propto w^{|H|+a-1}(1-w)^{|T|+b-1}$$
$$\propto \mathrm{Beta}(w||H|+a, |T|+b)$$

Bayesian Approach: Prediction

Prediction problem:

How can we predict the probability that next coin toss is Head, given observation D using Bayesian approach:

$$\boldsymbol{p(H|D)}$$

A toss depends on w, but we don't know what specific value for w we should use. So integrate over all possible value of w.

$$p(H|D) = \int_0^1 p(H, w|D)dw$$

Bayesian Approach: Prediction

Probability of head given data

$$p(H|D) = \int_0^1 p(H, w|D)dw$$ sum rule

$$= \int_0^1 p(H|w, D)p(w|D)dw$$ chain rule

$$= \int_0^1 w\, p(w|D)dw$$ def of model

$$= \mathbf{E}[w|D]$$ def of expectation

$$= \frac{|H| + a}{|H| + |T| + a + b}$$ from Beta distribution

Bayesian Approach: Prediction

This shows what?

We can predict the class of a new sample (e.g. what will be the result of a new coin toss) based on the prior and the data.

$$p(H|D) = \frac{|H| + a}{|H| + |T| + a + b}$$

- a = b = 100: A strong prior belief that the coin is fair
- a = b = 2: A weak prior belief that the coin is fair
- a=100, b=1: For a coin biased towards heads

- If D = {T, T} , probability that next coin is H?

Summary 1/3: Probability

- Random variables X take on values x from some domain D_X
- Probability distributions of X given by mass or density functions p
- Probabilities (masses) lie in $[0,1]$ and sum to one $\sum_{x \in D_x} p(x) = 1$
- Densities integrate to one $\int_{-\infty}^{\infty} p(x)dx = 1$
- Conditional probability $p(y|x)$
- Joint probability / product rule $p(x,y) = p(y|x)p(x)$
- Marginal probability $p(x) = \sum_{x \in D_x} p(x,y)$ or $p(x) = \int_{D_x} p(x,y)dx$
- Bayes' theorem $p(x|y) = p(y|x)p(x)/p(y)$
- Expected value $E[X]$ (if $D_X \subseteq \mathbb{R}$) $\sum_{x \in D_x} x\, p(x)$ or $\int_{D_X} x\, p(x)dx$
- Conditional expectations $E[Y \mid X = x] = \int_{D_Y} y\, p(x,y)dy$

Summary 2/3: Uncertainty

- Probabilistic models describe conditional distribution of target variable given input variables (and parameters) and thus describe aleatoric uncertainty of modelled process
- Maximum Likelihood Estimation (MLE): choose parameter values to maximise (modelled) probability of observed data
- Typically done by minimising the negative log likelihood
- Epistemic uncertainty about parameters and predictions due to finite data, i.e., could have observed different X and also different Y given X (noise)

Summary 3/3: Quantifying Uncertainty

- Bootstrap quantifies epistemic uncertainty by re-fitting model with datasets re-sampled from available data
- Results in distribution over parameters, predictions, and performance estimates
- Bayesian models instead define directly prior probability distributions on parameters
- Results in posterior distribution (given the data) of parameters and predictions that also quantifies epistemic uncertainty (use Bayes' theorem to evaluate posterior)

Laboratory (Week 2)

- Activity 1.2: bootstrap technique

  We will implement the Bootstrap technique to assess variations in the prediction of KNN classifier

- Activity 0.2: theory of continuous probability distributions and maximum likelihood estimate

  Attempt in your own time to consolidate material

What will we learn in Week 3

- **Part A in Module 2**
  - Linear Basis Functions
  - Optimising Error Functions
  - Regularisation algorithms

THANK YOU
for your attention!